

Finite-smoothness additivity of quasi-states on surfaces

Candidate full answer to Problem 8.2 of arXiv:math/0410338

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Status. Candidate full answer, likely valid, pending independent expert and novelty review. In fact the result holds for Lipschitz functions whose differentials are dependent almost everywhere, which is stronger than the requested finite-smoothness extension.

1 Question and answer

Entov and Polterovich prove strong quasi-additivity on a closed symplectic surface for smooth Poisson-commuting functions. They then ask whether the identity

$$\zeta(F + G) = \zeta(F) + \zeta(G) \tag{1}$$

extends to Poisson-commuting functions of finite smoothness [1, Problem 8.2].

Problem 8.2. *Extend identity (9) to Poisson-commuting functions of finite smoothness.*

For instance, one can try to find a uniform approximation of the pair (F, G) by a Poisson-commuting pair of C^∞ -functions.

Here a quasi-state means an Aarnes quasi-state on $C(M)$: a normalized positive functional whose restriction to the closed unital algebra generated by any one function is linear. No symplectic invariance is needed in the argument below.

Theorem 1 (Lipschitz additivity). *Let M be a closed surface and let ζ be any quasi-state on $C(M)$. If $F, G \in \text{Lip}(M)$ and*

$$dF \wedge dG = 0 \quad \text{almost everywhere on } M, \tag{2}$$

then (1) holds.

Corollary 2 (Full finite-smoothness answer). *Let (M, ω) be a closed symplectic surface. If $F, G \in C^1(M)$ and $\{F, G\} = 0$, then every quasi-state on $C(M)$ satisfies (1). In particular this holds for the quasi-state associated to every symplectic quasi-measure in the source paper.*

Indeed, on a symplectic surface $dF \wedge dG = \{F, G\}\omega$ (up to the harmless sign convention), and every C^1 function on compact M is Lipschitz. Thus Corollary 2 answers Problem 8.2 at the minimal classical differentiability at which the Poisson bracket is pointwise defined.

2 Proof

The source proof studies the joint image

$$\Phi : M \longrightarrow \mathbb{R}^2, \quad \Phi(x) = (F(x), G(x)), \quad \Delta := \Phi(M),$$

and obtains $\dim_{\text{cov}} \Delta \leq 1$ from a smooth Sard-type theorem. At Lipschitz regularity, the area formula and elementary dimension theory give the same topological conclusion more directly.

Lemma 3 (The joint image is at most one-dimensional). *Under the hypotheses of Theorem 1, $\dim_{\text{cov}} \Delta \leq 1$.*

Proof. Choose finitely many coordinate patches and compact subsets of them covering M . In each chart, Φ is a Lipschitz map between planar sets. Its approximate Jacobian is

$$J_2\Phi = |\det D(F, G)|,$$

which vanishes almost everywhere by (2). The Lipschitz area formula therefore gives, on every such chart domain E ,

$$\int_{\mathbb{R}^2} N(\Phi, E, y) dy = \int_E J_2\Phi(x) dx = 0.$$

Since $N(\Phi, E, y) \geq 1$ on $\Phi(E)$, every chart image has planar Lebesgue measure zero. Hence $\mathcal{L}^2(\Delta) = 0$.

The set Δ is compact, because M is compact and Φ is continuous. It has empty interior because it is planar-null. The classical Euclidean dimension theorem says that a subset of $\mathbb{R}^2$ has covering dimension 2 if and only if it contains an interior point; equivalently, every subset with empty interior has covering dimension at most 1 [3, Theorem 1.8.10]. Thus $\dim_{\text{cov}} \Delta \leq 1$. $\square$

Proof of Theorem 1. Define

$$\eta : C(\Delta) \longrightarrow \mathbb{R}, \quad \eta(h) = \zeta(h \circ \Phi).$$

This is a quasi-state. Positivity and normalization are immediate. If $h \in C(\Delta)$, pullback sends the singly generated algebra $\{\varphi \circ h : \varphi \in C(h(\Delta))\}$ into the singly generated algebra of $h \circ \Phi$, on which ζ is linear.

By the Wheeler–Shakhmatov theorem used in the proof of [1, Theorem 8.1], every quasi-state on a compact metric space of covering dimension at most one is linear. Hence η is linear. Apply linearity to the two coordinate functions p_1, p_2 on Δ :

$$\zeta(F + G) = \eta(p_1 + p_2) = \eta(p_1) + \eta(p_2) = \zeta(F) + \zeta(G).$$

This proves the theorem. $\square$

3 Checks, scope, and novelty status

There is no hidden injectivity assumption in the measure step: the area formula counts multiplicity, and zero Jacobian forces the multiplicity integral—hence the entire image measure—to vanish. Nor is Hausdorff dimension being inferred from measure zero; the proof uses the separate topological theorem that a planar subset of full covering dimension must have interior. This distinction is essential because planar-null sets can have Hausdorff dimension two.

The Lipschitz statement is stronger than requested and avoids the source’s suggested simultaneous smooth approximation problem. For C^1 functions, Poisson commutation gives (2) everywhere. At the Lipschitz level the natural invariant formulation is exactly (2) almost everywhere, using Rademacher differentials. Purely continuous functions have no intrinsic Poisson bracket, so no still-lower regularity is claimed.

Exact-title, exact-problem-phrase, “finite smoothness”, Wheeler–Shakhmatov, area-formula, and Poisson-commuting quasi-state searches through 11 August 2026 found the source problem and later neighboring work on C^0 Poisson-bracket rigidity, but no paper stating this resolution. The proof should therefore be treated as a candidate new observation until an expert completes a citation and novelty audit.

References

- [1] M. Entov and L. Polterovich, *Quasi-states and symplectic intersections*, arXiv:math/0410338v2; Comment. Math. Helv. 81 (2006), 75–99.
- [2] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, revised edition, CRC Press, 2015. See the Lipschitz area formula.
- [3] R. Engelking, *Dimension Theory*, North-Holland, 1978.